

A Study on Diffusion Models

Diffusion Models from the scratch



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Outline

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How can we define Ito Integral?

05 Ito Formula

How can we calculate Ito Integral?

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Overview

What we learn, and why we learn it

Preliminaries

Before we start SDE

SDE

$$\begin{aligned}dX_t &= b(X_t, t)dt + \sigma(X_t, t)dW_t \\&\downarrow \\X_t &= X_0 + \int_0^t b(X_s, s)ds + \int_0^t \sigma(X_s, s)dW_s\end{aligned}$$

- What is the **stochastic process** $\{X_t\}_{t \geq 0}$? (X_t 's are **random variables**).
- How can we define **Ito Integral** $I(t) = \int_0^t \sigma(X_s, s)dW_s$?
- How can we calculate Ito Integral? (**Ito Formula**)

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Basic Measure Theory

How can we measure a set?

Basic Measure Theory

Why we need?



Problem

반지름이 5인 원판에 다트를 던지는 상황

■ Sample Space :

$$\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 25\}$$

■ Event : $\mathcal{F} = \mathcal{P}(\Omega)$

■ Probability Measure : $P(A) = \frac{\text{Area of } A}{25\pi}$

But, 모든 $A \in \mathcal{F}$ 에 대해서 좋은 성질을 만족하는 P 는 존재하지 않는다 (Vitali Set).

TODO

따라서 Measurable set을 정의하고, 이후 Measure를 정의해야한다.

Basic Measure Theory σ -algebra and measure

σ -algebra

$X \neq \emptyset, \mathcal{M} \subseteq \mathcal{P}(X)$. $\mathcal{M}$ is a **σ -algebra** if it is closed under countable union and complement.

$\Rightarrow (X, \mathcal{M})$: **measurable space**, $E \in \mathcal{M}$: **measurable set**

Measure

$\mu : \mathcal{M} \rightarrow [0, \infty]$ such that

$$1 \quad \mu(\emptyset) = 0$$

$$2 \quad \mu\left(\bigsqcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i)$$

$\Rightarrow \mu$: **measure**, $(X, \mathcal{M}, \mu)$: **measure space**

Generated σ -algebra

For $E \in \mathcal{P}(X)$, $\sigma(E)$ is the **σ -algebra generated by E** :

$$\sigma(E) = \bigcap_{\substack{E \subseteq \mathcal{M} \\ \mathcal{M}: \sigma\text{-alg.}}} \mathcal{M}$$

= the smallest σ -algebra containing E .

Summary

$$\begin{array}{l} X : \text{a set} \\ \downarrow + \sigma\text{-alg.} \\ (X, \mathcal{M}) : \text{measurable space} \\ \downarrow + \text{meas.} \\ (X, \mathcal{M}, \mu) : \text{measure space} \end{array}$$

Basic Measure Theory

Measurable functions and Borel measure

Measurable function

Let $(X, \mathcal{M})$, $(Y, \mathcal{N})$ be measurable spaces. Then $f : X \rightarrow Y$ is $\mathcal{M}$ - $\mathcal{N}$ **measurable** if

$$f^{-1}(F) \in \mathcal{M} \quad \forall F \in \mathcal{N}.$$

If $Y = \mathbb{R}^n$ (topological space), we simply say f is $\mathcal{M}$ -measurable.

Borel measure

■ $\mathcal{B}$: Borel σ -algebra

■ μ : Borel measure

$\mathbb{R}^n$ 에서 가장 기본적인 measure.

c.f. completion $\rightarrow$ Lebesgue measure

c.f. Multi Dimension, Subset of $\mathbb{R}^n$

Borel σ -algebra on $\mathbb{R}$

$\mathcal{B}_{\mathbb{R}}$ is generated by one of

$$E_1 = \{(a, b) : a < b\},$$

$$E_3 = \{[a, b) : a < b\},$$

$$E_5 = \{(a, \infty) : a \in \mathbb{R}\},$$

$$E_2 = \{[a, b] : a < b\},$$

$$E_4 = \{(a, b] : a < b\},$$

$$E_6 = \{(-\infty, b) : b \in \mathbb{R}\}.$$

Borel measure

$F : \mathbb{R} \rightarrow \mathbb{R}$, right continuous $\Rightarrow \mu_F((a, b]) := F(b) - F(a)$

μ_F : Borel measure (the easiest one: $F = I$)

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Basic Probability Theory

What is the probability of a set?

Basic Probability Theory

[Go back to the example](#)

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Borel Sets

- $\mathcal{F}$ 를 σ -algebra로 만들기 : $\mathcal{F} = \mathcal{B}_\Omega$
- P 를 $\mathcal{F}$ 의 measure로 만들기 : $P = \mu_I$

Basic Probability Theory

Probability space and random variable

Definition: probability space

$(\Omega, \mathcal{F}, P)$ is a **probability space** if

- Ω : sample space (set of outcomes)
- $\mathcal{F}$: σ -algebra
- P : probability measure

and $A \in \mathcal{F}$ is called an **event**.

Definition: random variable

X is a **random variable** of $(\Omega, \mathcal{F}, P)$ if

$$X : \Omega \rightarrow \mathbb{R}, \quad \mathcal{F}\text{-measurable function.}$$

And if X_t 's are random variables, then $\{X_t\}_{t \geq 0}$ is a **stochastic process**.

Basic Probability Theory

Example: Probability space and random variable

Example: Probability space and random variable

반지름이 5인 원판에 다트를 던지는 상황

- Sample Space : $\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 25\}$
- Event : $\mathcal{F} = \mathcal{B}_\Omega$
- Probability Measure : $P(A) = \mu_I(A) \quad (= \frac{\text{Area of } A}{25\pi})$

이때, 함수 $X : \Omega \rightarrow \mathbb{R}$ 를 다음과 같이 정의하자:

$$X(x, y) = \sqrt{x^2 + y^2} \quad \text{과녁 중앙에서부터 다트까지의 거리}$$

그러면 X 는 $\mathcal{F}$ -measurable이므로 random variable이다.

$$\begin{aligned} P(1 \leq X \leq 3) &= P(X^{-1}([1, 3])) \\ &= P(\{(x, y) \in \Omega : 1 \leq \sqrt{x^2 + y^2} \leq 3\}) \\ &= \frac{9\pi - \pi}{25\pi} = \frac{8}{25} \end{aligned}$$

Basic Probability Theory

Important properties of Brownian motion

Important properties of Brownian motion

Brownian Motion B_t is a stochastic process that satisfies the following properties:

1 B_t is a **Gaussian process**, that is

$$(B_{t_1}, \dots, B_{t_k}) \sim \mathcal{N} \quad \text{for all } t_i.$$

2 B_t has **independent increments**, that is

$$B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_k} - B_{t_{k-1}}$$

are independent for all $0 \leq t_1 < \dots < t_k \leq T$.

3 There exists a **continuous version** of B_t if Kolmogorov's condition holds.

Furthermore, by definition (usually $x = 0$)

- $B_0 \sim \mathcal{N}(x, t)$
- $B_t - B_s \sim \mathcal{N}(0, t - s)$ for all $0 \leq s < t \leq T$.
- $\mathbb{E}[(B_t - x)(B_s - x)] = \min(t, s)$

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Ito Integral

How can we define Ito Integral?

Ito Integral

Construction of Ito Integral

Construction of Ito Integral

Let $\mathcal{V}$ be a function class that we defined in the note. We construct the Ito Integral on this function class.

1. Characteristic Function :

$$\chi_{[t_j, t_{j+1}]}(t)$$

2. Elementary Function :

$$\sum_{j=1}^n e_j(w) \chi_{[t_j, t_{j+1}]}(t)$$

3. Bounded Continuous Function in $\mathcal{V}$:

$$g(\cdot, w)$$

4. Bounded Function in $\mathcal{V}$:

$$h$$

5. Function in $\mathcal{V}$:

$$f$$

where $\int_S^T e_j(w) \chi_{[t_j, t_{j+1}]}(t) dB_t = e_j(w)(B_{t_{j+1}} - B_{t_j})$ for all $S \leq t_j \leq t_{j+1} \leq T$.

Ito Integral

Ito Isometry

The Ito Isometry

$$\mathbb{E} \left[\left(\int_S^T f dB_t \right)^2 \right] = \mathbb{E} \left[\int_S^T f^2 dt \right] \quad \text{for all } f \in \mathcal{V}(S, T).$$

Example: second moment of an Ito integral

Closed form 없이 $\mathbb{E}[(\int_0^T B_t dB_t)^2]$ 를 구하고 싶을 때, Ito isometry가 stochastic integral을 Lebesgue integral로 바꿔 준다.

$$\mathbb{E} \left[\left(\int_0^T B_t dB_t \right)^2 \right] = \mathbb{E} \left[\int_0^T B_t^2 dt \right] = \int_0^T \mathbb{E}[B_t^2] dt = \int_0^T t dt = \frac{T^2}{2}.$$

Ito Integral

Multi-Dimensional Ito Integral

From one dimension to n dimensions

$$\int f dB \ (\mathbb{R} \rightarrow \mathbb{R}, \mathcal{V}) \rightarrow \sum_{j=1}^n \int v_j dB^j \ (\mathbb{R}^n \rightarrow \mathbb{R}, \mathcal{V}^{1 \times n}) \rightarrow \left(\sum_{j=1}^n \int v_{ij} dB^j \right)_{i=1}^m \ (\mathbb{R}^n \rightarrow \mathbb{R}^m, \mathcal{V}^{m \times n})$$

Definition

$$v = \begin{pmatrix} v_{11} & \cdots & v_{1n} \\ \vdots & \ddots & \vdots \\ v_{m1} & \cdots & v_{mn} \end{pmatrix}, \quad B = \begin{pmatrix} B_1 \\ \vdots \\ B_n \end{pmatrix}$$

$$\int_S^T v dB = \begin{pmatrix} \int_S^T v_1 dB \\ \vdots \\ \int_S^T v_m dB \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^n \int_S^T v_{1j} dB_j \\ \vdots \\ \sum_{j=1}^n \int_S^T v_{mj} dB_j \end{pmatrix}$$

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Ito Formula

How can we calculate Ito Integral?

Ito Formula

Why Ito Formula?

Ordinary calculus

- Fundamental Theorem of Calculus:

$$F(t) = \int_0^t f'(y) dy$$

- Chain rule: $\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x_t} \cdot \frac{dx_t}{dt}$

Ito integral에도 이런 좋은 방법이 있을까?

Ito formula (roughly)

For $Y_t = g(t, X_t)$,

$$\frac{dY_t}{dt} = \frac{\partial g}{\partial t} + \frac{\partial g}{\partial x_t} \frac{dx_t}{dt} + \frac{1}{2} \frac{\partial^2 g}{\partial x_t^2} \frac{(dx_t)^2}{dt}.$$

Ito Formula

Ito Formula

Definition: one-dimensional Ito process

Let $(\Omega, \mathcal{F}, P)$ be a probability space and B_t a one-dimensional Brownian motion. X_t is an **Ito process** if

$$X_t = X_0 + \int_0^t u(s, \omega) ds + \int_0^t v(s, \omega) dB_s, \quad dX_t = u dt + v dB_t.$$

Ito formula

Let X_t be an Ito process ($dX_t = u dt + v dB_t$) and $g(t, x) \in C^2([0, \infty) \times \mathbb{R})$. Then $Y_t := g(t, X_t)$ is also an Ito process, and

$$dY_t = \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} (dX_t)^2.$$

Computing rules: $(dt)^2 = dt \cdot dB_t = dB_t \cdot dt = 0$, $(dB_t)^2 = dt$.

Ito Formula

Example: Ito Formula

Examples

1 $\int_0^t dB_s = B_t$

2 $\int_0^t B_s dB_s = \frac{1}{2}(B_t^2 - t)$

$g(x) = x^2$: $B_t^2 = 2 \int_0^t B_s dB_s + \int_0^t (dB_s)^2$, and $(dB_s)^2 = ds$.

3 $\int_0^t s dB_s = tB_t - \int_0^t B_s ds$
 $g(t, x) = tx$.

Integration by parts

If $f(t, \omega) = f(t)$ (deterministic), then

$$\int_0^t f(s) dB_s = f(t)B_t - \int_0^t B_s df_s.$$

Ito Formula

Multi-Dimensional Ito Formula

Definition: n -dimensional Ito process

$dX = u dt + v dB$, where $X(t), u$ are $n \times 1$, $B(t)$ is $m \times 1$, and v is $n \times m$:

$$dX_i(t) = u_i dt + \sum_{j=1}^m v_{ij} dB_j.$$

n -dimensional Ito formula

For $g \in C^2([0, \infty) \times \mathbb{R}^n, \mathbb{R}^p)$,

$$dY_k = \frac{\partial g_k}{\partial t} dt + \sum_i \frac{\partial g_k}{\partial x_i} dX_i + \frac{1}{2} \sum_{i,j} \frac{\partial^2 g_k}{\partial x_i \partial x_j} dX_i dX_j.$$

Computing rules: $dB_i dB_j = \delta_{ij} dt$, $(dt)^2 = dt \cdot dB_i = dB_i \cdot dt = 0$.

Thank you



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